**REVISION WORKSHEET**

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

# AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

## Practice Questions and Worked Answers

**Grade 11 Advanced | Module 9 Lesson 1 | Impulse and Momentum**

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: $p = mv$; $J = F_{avg}\Delta t = \Delta p$; $L = I\omega$; $\tau_{net}\Delta t = \Delta L$. Use signed velocities and consistent directions.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 11 book, printed pages 30–37.

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**P01. Linear momentum is:**

A. Mass divided by speed  B. Mass multiplied by velocity
C. Force divided by time  D. Energy multiplied by distance

**Answer**

B. Mass multiplied by velocity.

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**P02. The area under a force–time graph represents:**

A. Velocity  B. Position
C. Impulse  D. Power

**Answer**

C. Impulse.

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**P03. For a fixed change in momentum, increasing stopping time reduces:**

A. Impulse magnitude  B. Mass necessarily
C. Change in velocity necessarily  D. Average stopping force

**Answer**

D. Average stopping force.

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**P04. Angular momentum about a fixed axis is:**

A. $I\omega$  B. $I/\omega$
C. $\tau/\omega$  D. $mr$

**Answer**

A. $I\omega$.

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**P05. The SI unit N s is equivalent to:**

A. kg $m^2/s^2$  B. kg m/s
C. kg/s  D. $m/s^2$

**Answer**

B. kg m/s.

**P06. A 1 kg object moves at +4.0 m/s. Calculate its linear momentum.**

A. 8 kg m/s  B. 2 kg m/s
C. 4 kg m/s  D. -4 kg m/s

**Answer**

C. 4 kg m/s. $p = mv = 1 \times 4.0 = +4$ kg m/s.

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**P07. A constant force +40 N acts for 0.050 s. Find its impulse.**

A. 4 N s  B. 1 N s
C. -2 N s  D. 2 N s

**Answer**

D. 2 N s. $J = F\Delta t = 40 \times 0.050 = +2$ N s.

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**P08. A 0.4 kg ball changes velocity from −3.0 to +5.0 m/s in 0.040 s. Find its average force.**

A. 80 N  B. 160 N
C. 40 N  D. -80 N

**Answer**

A. 80 N. $F_{avg} = m(v_f - v_i)/\Delta t = 0.4(5.0 - (-3.0))/0.040 = +80$ N.

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**P09. A rotor has I = 0.4 kg m² and angular velocity +6.0 rad/s. Find its angular momentum.**

A. 4.8 kg m²/s  B. 2.4 kg m²/s
C. 1.2 kg m²/s  D. -2.4 kg m²/s

**Answer**

B. 2.4 kg m²/s. $L = I\omega = 0.4 \times 6.0 = +2.4$ kg m²/s.

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**P10. A constant net torque +6 N m acts for 0.20 s. Find the change in angular momentum.**

A. 2.4 kg m²/s  B. 0.6 kg m²/s
C. 1.2 kg m²/s  D. -1.2 kg m²/s

**Answer**

C. 1.2 kg m²/s. $\Delta L = \tau\Delta t = 6 \times 0.20 = +1.2$ kg m²/s.

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**P11. Define impulse.**

**Answer**

The change in momentum; for constant force it is force multiplied by the time interval, and more generally it is the force–time integral.

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**P12. State the impulse–momentum theorem.**

**Answer**

Net impulse on an object equals the change in its momentum: $J_{net} = \Delta p$.

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**P13. Define angular momentum for rotation about a fixed axis.**

**Answer**

$L = I\omega$, with units kg m²/s, for the rigid-axis situation described.

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**P14. Why can airbags reduce injury in a crash?**

**Answer**

They increase the time over which a person's momentum changes and distribute contact forces, reducing typical average forces for a given stopping impulse.

**P15. State the angular impulse–angular momentum theorem.**

**Answer**

The net torque integrated over time equals the change in angular momentum; for constant net torque, $\tau_{net}\Delta t = \Delta L$.

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**P16. A 1.5 kg object moves at +4.0 m/s. Calculate its linear momentum.**

**Answer**

$p = mv = 1.5 \times 4.0 = +6$ kg m/s.

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**P17. A 2 kg object moves at +4.0 m/s. Calculate its linear momentum.**

**Answer**

$p = mv = 2 \times 4.0 = +8$ kg m/s.

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**P18. A constant force +60 N acts for 0.050 s. Find its impulse.**

**Answer**

$J = F\Delta t = 60 \times 0.050 = +3$ N s.

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**P19. A constant force +80 N acts for 0.050 s. Find its impulse.**

**Answer**

$J = F\Delta t = 80 \times 0.050 = +4$ N s.

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**P20. A 0.6 kg ball changes velocity from −3.0 to +5.0 m/s in 0.040 s. Find its average force.**

**Answer**

$F_{avg} = m(v_f - v_i)/\Delta t = 0.6(5.0 - (-3.0))/0.040 = +120$ N.

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**P21. A 0.8 kg ball changes velocity from −3.0 to +5.0 m/s in 0.040 s. Find its average force.**

**Answer**

$F_{avg} = m(v_f - v_i)/\Delta t = 0.8(5.0 - (-3.0))/0.040 = +160$ N.

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**P22. A rotor has I = 0.6 kg m² and angular velocity +6.0 rad/s. Find its angular momentum.**

**Answer**

$L = I\omega = 0.6 \times 6.0 = +3.6$ kg m²/s.

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**P23. A rotor has I = 0.8 kg m² and angular velocity +6.0 rad/s. Find its angular momentum.**

**Answer**

$L = I\omega = 0.8 \times 6.0 = +4.8$ kg m²/s.

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**P24. A constant net torque +9 N m acts for 0.20 s. Find the change in angular momentum.**

**Answer**

$\Delta L = \tau\Delta t = 9 \times 0.20 = +1.8$ kg m²/s.

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**P25. A constant net torque +12 N m acts for 0.20 s. Find the change in angular momentum.**

**Answer**

$\Delta L = \tau\Delta t = 12 \times 0.20 = +2.4$ kg m²/s.

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**P26. A force–time graph is triangular with base 0.020 s and peak force 300 N. Calculate impulse.**

**Answer**

Impulse equals triangular area = ½ × 0.020 × 300 = 3.0 N s in the force direction.

**P27. A ball moving toward a bat rebounds in the opposite direction. Why must signs be used in calculating Δp?**

**Answer**

The final and initial velocities have opposite signs. Subtracting signed momenta includes the full reversal and avoids underestimating the impulse.

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**P28. Compare catching a ball with rigid hands and with hands moving backward. Which usually reduces force and why?**

**Answer**

Moving the hands backward increases stopping time. For the same momentum change, $F_{avg} = \Delta p/\Delta t$ is smaller.

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**P29. An ice skater pulls in their arms with negligible external torque. Explain what happens to angular speed.**

**Answer**

Angular momentum stays constant. Pulling in the arms reduces I, so ω increases to keep Iω constant.

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**P30. A large force acts for a very short time. Is its impulse necessarily greater than a small force over a long time?**

**Answer**

No. Impulse depends on the force–time area, not force alone. Either case can have the greater impulse depending on both force and duration.